

C. Balkı GEMİRTER

Chief AI Officer (CAIO) & R&D Leadership

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Location İstanbul, Türkiye **Languages** Turkish (Native) | English (Professional) | Japanese (Elementary)

Artificial Intelligence & Deep Learning | HealthTech Solutions | FinTech & Predictive Analytics | R&D & Innovation Management | Large Language Models (LLM) & NLP | IoT & Smart City Analytics

19+ YEARS OF EXPERIENCE | **6** PUBLICATIONS | **9** R&D PROJECTS | **16** CERTIFICATES

PROFILE

Ph.D. candidate in Computer Engineering at Yeditepe University (4.00 GPA) with 19+ years of enterprise R&D leadership and deep-tech entrepreneurship experience. Built upon a strong corporate engineering foundation at TEB and Nortel/Netaş, with concurrent leadership in 0-to-1 AI architectures, TÜBİTAK-funded projects, and LLM implementations as CAIO and AI Director across high-growth startups (CEIBA Health, Bayegan, Agada Tech, Vivoo). Specialized in transforming advanced AI research into scalable products and measurable business value in healthcare, finance, and commodity trading.

EXPERIENCE

STARTUP & TRANSFORMATION LEADERSHIP — 2023 – Present

AI strategy and product architecture build-outs at deep-tech startups, carried out concurrently with doctoral research.

CEIBA HEALTH — Jan 2026 - Apr 2026

Director of Artificial Intelligence

- Led AI and R&D strategy for next-generation healthcare solutions
- Secured and managed TÜBİTAK 1501 and 1832 projects
- Directed YOLO-based real-time fall risk detection (>90% accuracy, <10s alert latency) and architected both the LLM-powered "AI Companion" for automated SOAP notes and a DSP-based strip analysis system for ECG rhythm analysis

Bayegan — May 2025 - Jul 2025

Chief AI Officer (CAIO)

- Spearheaded the AI roadmap of a commodity trading company, aligning it with business objectives to enhance predictive analytics and risk management
- Directed model training and optimization for automated trading systems
- Fostered collaboration with trading desks and software engineers to integrate AI into workflows

Agada Tech — Sep 2024 - Jan 2025

VP of Artificial Intelligence

- Spearheaded AI-driven customer segmentation models and oversaw real-time next-order prediction systems for high-volume e-commerce
- Pioneered advanced AI methodologies to enhance customer loyalty strategies

Yapı Kredi Teknoloji — May 2024 - Aug 2024

AI & Applied Data Science Manager

- Led a team of 17 engineers with 0% attrition
- Managed critical AI projects such as Digital Mizan and Fraud Detection
- Delivered cost and FTE savings through computer vision, anomaly detection, and deep learning
- Led ITU Teknopark-supported R&D and contributions to Horizon 2020

Vivoo — Aug 2023 - Feb 2024

Chief AI Officer (CAIO) & Data Analytics

- Managed the development of "Welly", an LLM-based wellness assistant, and AI-driven test strip analysis tools
- Led a TÜBİTAK 1501-supported project analyzing nutritional values from meal photos

ENTERPRISE R&D & SOFTWARE MANAGEMENT — 2007 – 2023

R&D, team leadership, and patent/TÜBİTAK processes within large-scale banking and telecommunications infrastructures.

TEB - Türk Ekonomi Bankası — 4 roles — Jan 2013 - Aug 2023

Senior R&D Manager (Artificial Intelligence, Software) — Aug 2016 - Aug 2023

- Led a team of 10 engineers in TEB's R&D department
- Idea owner and supervisor of the TÜBİTAK TEYDEB 1501-supported NEURON (AI chatbot) and ConnecTEB projects
- Led deep learning/NLP initiatives such as Dragon and Ribbon
- Pioneered the transition of the first R&D team to TEB ARF

Infrastructure Software Development Assistant Project Leader
Oct 2015 - Aug 2016

- Architect of the core IT framework used by TEB's software designers across all banking projects
- Led scalability, performance, and new-requirement integration efforts

Switch and ATM Software Development Assistant Project Leader
Nov 2014 - Oct 2015

- Core member of the Aquila team on TEB's first TÜBİTAK TEYDEB 1501 project, a 100% in-house XTM system
- Led a team of 5 engineers from prototype to full-scale deployment

Internet Banking Software Development Project Supervisor
Jan 2013 - Nov 2014

- Team lead for over 10 engineers across retail and commercial internet banking
- Spearheaded the redesign of both platforms and delivered an advanced smart search feature

NORTEL Networks - NETAŞ
2 roles | Jul 2007 - Jan 2013

Senior Software Design Engineer
Mar 2010 - Jan 2013

- Designed and developed the A2 Base/OAM system manager and its sub-components using Java and Oracle
- Built high-performance, resilient architectures for critical network infrastructure

Junior Software Design Engineer
Jul 2007 - Mar 2010

- Developed and tested system management (OAM) software components in Java for telecommunications network infrastructure

EDUCATION

2020 - Nov 2026 (Expected)

Ph.D. - Computer Engineering
Yeditepe University | Taught in English

4.00 / 4.00 — Ranked 1st, Full Scholarship

THESIS
Incorporating prior knowledge in Deep Learning (DL) architectures for image recognition and medical diagnosis with small cohort data.

2018 - 2020

M.Sc. - Computer Engineering
Yeditepe University | Taught in English

3.87 / 4.00 — Ranked 1st

THESIS
A Turkish Question Answering System Based on Deep Learning Networks (BERT).

2002 - 2007

B.Sc. - Computer Engineering
Ege University

3.34 / 4.00 — Ranked 4th

Education in core computer engineering disciplines and software architecture.

88 — **ALES** 2018
Academic Personnel and Postgraduate Education Entrance Exam

85 — **YÖKDİL** 2026
Higher Education Institutions Foreign Language Exam (English)

ACADEMIC PUBLICATIONS

01 **RP-Fuse: Development of an Adaptive Dose-Gated Multimodal Model for Predicting Radiation Pneumonitis in Non-Small Cell Lung Cancer** SCIE - Q2
MDPI - Applied Sciences (2026) | C. Balkı GEMİRTER, Esra KORKMAZ KIRAKLI, Cemre Ceren ÖZDİLLİ, Ali Kadri ÇIRAK, E. Erkan KORKMAZ, Dionysis GOULARAS

02 **CoDA: A Cognitive-Inspired Approach for Domain Adaptation** SCIE - Q2
MDPI - Applied Sciences (2026) | C. Balkı GEMİRTER, E. Erkan KORKMAZ, Dionysis GOULARAS

03 **Location Proofing Using Video Similarity** TR Dizin
Journal of Scientific Reports-A (2022) | C. Balkı GEMİRTER, Tacha SERIF

04 **A Turkish Question Answering System Based on Deep Learning Neural Networks** TR Dizin
JISTA - Journal of Intelligent Systems (2021) | C. Balkı GEMİRTER, Dionysis GOULARAS

05 **Fine-Tuning Throughput and QoS on SMT Cores** IEEE
UBMK 2021 - 6th International Conference on Computer Science and Engineering (2021) | C. Balkı GEMİRTER, Gürhan KÜÇÜK

06 **A Comparative Evaluation of AMQP, MQTT and HTTP Protocols Using Real-Time Public Smart City Data** IEEE
UBMK 2021 - 6th International Conference on Computer Science and Engineering (2021) | C. Balkı GEMİRTER, Çağatay ŞENTÜRCA, Şebnem BAYDERE

TÜBİTAK & R&D PROJECTS

TÜBİTAK TEYDEB 1501 — 2026

Image Processing-Supported Patient Monitoring and Fall Risk Management
CEIBA HEALTH
Supervised R&D project focusing on AI-based healthcare solutions.

TÜBİTAK TEYDEB 1832 — 2026

Green and Sustainable Digital Solutions in the Health Sector
CEIBA HEALTH
Supervised R&D initiative for eco-friendly digital health.

TÜBİTAK TEYDEB 1501 — 2024 – 2025

AI-Based Dynamic Customer Segmentation and Personalized Gamification System
Agada Tech
Advanced AI models for user retention and personalized experiences.

TÜBİTAK TEYDEB 1501 — 2023 – 2024

AI-Based Nutrition Analysis System

Vivoo

Multimodal health AI project expanding personalized nutrition insights.

Boğaziçi Technopark — 2022 – 2023

Dragon - Banking Question Answering (MRQA) System

TEB - Türk Ekonomi Bankası

Machine reading comprehension-based smart search over banking documents, giving call center staff instant, concise answers to banking queries.

TÜBİTAK TEYDEB 1501 — 2020 – 2022

AI-Based Customer Chatbot

TEB - Türk Ekonomi Bankası

Developed NEURON, an AI chatbot integrated into digital banking.

TÜBİTAK TEYDEB 1501 — 2016 – 2018

ConnecTEB - VoIP & Multimedia Solution

TEB - Türk Ekonomi Bankası

In-house secure video conferencing tool adopted bank-wide.

TÜBİTAK TEYDEB 1501 — 2014 – 2016

Orion & Aquila (In-House XTM)

TEB - Türk Ekonomi Bankası

Core IT framework development and scaling of ATM/Switch infrastructure.

EU-Supported

InVisIBLe Plus

Yeditepe University

Inclusive and innovative learning tool for visually impaired and blind people.

ACADEMIC & PROFESSIONAL SERVICE

Turkish Journal of Electrical Engineering & Computer Sciences

Reviewer — TÜBİTAK

European Journal of Radiology

Reviewer — Elsevier

Bilişim Vadisi (Informatics Valley)

Mentor — Technopark

Yeditepe University

Teaching Assistant — 2021 – 2026

Academic assistantship and project consulting for Artificial Intelligence (CSE 462, AI 401), Deep Learning (CSE 588), and Algorithm Analysis (CSE 311) courses.

CERTIFICATIONS & AWARDS

AWARD — **Leaders of Tomorrow (x2)**

Global Talent Award · 2023 - 2024

Scrum.org — **PSPO-I Professional Scrum Product Owner**

Mar 2020

Scrum.org — **PSM-I Professional Scrum Master**

Feb 2017

Generative AI for Everyone — Mar 2025
DeepLearning.AI

Sentiment Analysis with Deep Learning using BERT — Feb 2022
Coursera

Deep Neural Networks with PyTorch — Aug 2021
IBM

Natural Language Processing Specialization — Dec 2020
DeepLearning.AI

Machine Learning — May 2020
Stanford Online

TensorFlow in Practice Specialization — Mar 2020
DeepLearning.AI

AI for Medicine Specialization — Jul 2022
DeepLearning.AI

Scalable Machine Learning on Big Data using Apache Spark — Nov 2021
IBM

Generative Adversarial Networks (GANs) Specialization — Feb 2021
DeepLearning.AI

Fundamentals of Parallelism on Intel Architecture — Nov 2020
Intel

Deep Learning — Apr 2020
DeepLearning.AI